HealthConnect Rwanda

Final Analytics and Decision Support Package (Week 8)

AnalystLab Africa Experience Lab, Data Analytics track | Yawa Jacqueline GBEDEVI | November 2026

1. From Week 7 to Week 8

Question	Answer
Main Week 7 output	Analytics Testing and Refinement Report, a 32-test validation suite (script 04), and three successive dashboard exports.
Most important testing result	Every financial total reproduces exactly (28 of 32 tests pass). Two Week 6 conclusions failed, since facility satisfaction and facility revenue rankings are not statistically distinguishable.
Major issue discovered	Only 5,000 of the 60,000 appointments referenced by billing exist, so facility, department and month views cover about 8% of bills.
What was refined	Date rule, outstanding amount, doctor workload query, department matrix relationships, payment method measure, decomposition tree label, complaint trend, rank display. The dashboard went from 14 of 20 checks passed to 21 of 23.
Component now ready	The database, the 40 queries, the validated KPI definitions, the reports and the dashboard, with two open dashboard items.
Still to integrate	Responses from Data Science and Project Management, which are not recorded yet.
Tracks involved	Data Science (findings that shape modelling decisions) and Project Management (issue log and presentation).
What I will demonstrate	The validated numbers, what survived testing, the decisions they support, and the limits of the data.

2. Final integration readiness

Requirement	Response
Final component	The analytics and decision-support layer: PostgreSQL database (raw, clean and analytics schemas), 40 SQL queries, validation suite, six-page Power BI dashboard, insights and testing reports, and presentation.
Problem addressed	Give HealthConnect managers validated numbers on patients, activity, revenue, collections, claims and staffing, and show where the data cannot yet support a decision.
Current status	Ready for integration. Open items are the province map, readability of dark matrix cells and one cosmetic label (ICU).
Improved in Week 7	A validation suite, statistical robustness checks, and three dashboard re-tests that fixed the defects they found, apart from three that stay open, and caught two problems caused by the fixes.
Depends on	The data owner for the source gaps, Data Science and Project Management for feedback on the findings handed over.
Tracks that depend on my output	Data Science (validated findings and data limits), Project Management (insights, issues, readiness), and potentially the assistant track (KPI definitions for executive questions).
Output I provide	Scripts 01 to 04, the validated KPI definitions below, six validated facts, eight logged issues, the dashboard and both reports.
Output I need	From Data Science, their prepared dataset or a summary to reconcile. From Project Management, confirmation that the issues are logged. Neither is recorded yet.
Evidence of readiness	28 of 32 automated tests (the four failures are documented), 21 of 23 dashboard checks, seven robustness tests, three dashboard exports.
Remaining limitations	8% appointment coverage, contradictory billing status, unexplained revenue summary, no admissions, laboratory, prescription or referral tables.

3. Executive summary

The objective of the project is to give HealthConnect Rwanda data-driven recommendations that improve service delivery, hospital operations, patient outcomes and financial sustainability. This package shows how the network operates and where money, capacity and data quality are at stake, and every number below has been recomputed from the raw data.

- **Collections are the clearest opportunity.** 9.31 billion was billed and 8.39 billion collected (90.1%). The remaining 920 million is spread over 11,887 bills, and the 90% rate is almost the same in every province and visit type.
- **Emergency load is high and uniform.** One visit in three is an emergency, in every facility and every month.
- **Facility rankings should not drive decisions yet.** Differences in satisfaction ($p = 0.79$) and in revenue between halves of the year are consistent with chance.
- **Data capture limits every conclusion.** Appointments and facility details are missing or inconsistent, which is the first thing to fix.

4. Final KPIs

KPI	Value	Basis	Validation
Registered patients	20,000	All patients	T01, D1
Average patient age	45 (range 0 to 90)	All patients	T25
Insurance coverage	100% (15 codes)	All patients	Query 6, caution on meaning
Appointments	5,000	File content	T02, D2
Emergency share	33.0%	5,000 appointments	B1, B2
Total billed	9,309,401,961	All 60,000 bills	T09, D3
Total collected	8,389,454,556	All bills	T10
Outstanding	919,947,405 (9.9%)	All bills	T11
Collection rate	90.1%	All bills	T12, B6
Average bill	155,157 (median 155,247)	All bills	Query 36
Claims and approval	42,036 claims, 50.0% approved	All claims	T18
Diagnoses, severe share	60,000, 33.3% severe	All diagnoses	D7, query 25
Doctors, total staff, facilities	350, 1,250, 20	All	T08, T19, D4
Average rating	2.99	21,064 reviews	T17, D5
Bed utilization	Not calculable	All beds Available	D14, documented

Revenue, appointments and ratings by facility or department cover only the bills, diagnoses and reviews linked to an appointment, about 8%.

5. Validated findings and where they appear

Finding	Evidence	Where shown
90% of billed amounts are collected, evenly across segments	T09 to T12, B6	Financial page, SQL query 35b
One visit in three is an emergency, stable everywhere	B1, B2	SQL query 16, executive report
Claims split evenly among four insurers, half still pending	T18, query 34	Financial page (claims by provider)
Patients balanced by gender, province and age	Queries 2, 3, 10, D6	Patient Analytics page
Five diagnoses of about 20% each, one third severe	Queries 21, 25, D7	Clinical Analytics page
Facility satisfaction and revenue rankings are not reliable	B3, B4, B5	Operational and Patient Experience pages, presented as descriptive
Facility-level views rest on 8% of bills	T29, D11	Blank facility row of the Operational matrix

6. Business insights and recommendations

Recommendation	Decision supported	Confidence	Evidence
Tighten collections on the 11,887 bills with a balance	Finance follow-up	High	T09 to T12, uniform 90% (B6)
Review the emergency pathway and record the reason for each emergency visit	Capacity planning	High	33% share, stable (B1, B2)
Set processing targets with the four insurers	Claims management	High	About half of the claims pending for all insurers
Match staffing to demand once full appointment history is loaded	Staffing	Conditional	Visit gap p = 0.052, 61 or 64 staff per facility
Load the full appointment history and add facility names and clinical tables	Planning and capacity decisions	High	Facility, department and month views rest on 8% of bills
Do not rank facilities on satisfaction or revenue yet	Performance management	High	B3 (p = 0.79), B4 (rank correlation 0.24)

The Week 6 business questions on admissions, laboratory demand, referrals, operating costs and regional investment cannot be answered, because the dataset has no such tables and no cost or location data. The contribution of this track to the project objective is therefore the validated analysis of what the data supports, and a clear list of what to capture next.

7. My part of the final walkthrough

- **The problem.** HealthConnect Rwanda needs data-driven recommendations on service delivery, hospital operations, patient outcomes and financial sustainability, and needs trustworthy data to decide where to act.
- **The data.** The Week 6 database, 27 tables, with 20,000 patients, 5,000 appointments, 60,000 bills, 42,036 claims, 60,000 diagnoses and 21,064 reviews. Raw data was never modified, and every correction is in a cleaned schema with a documented quality report.
- **The analytics.** Forty SQL queries, a validation suite, and a six-page dashboard. The data reveals even distributions almost everywhere, strong collections performance with a 10% gap, a high emergency share, and a data quality problem that limits facility comparisons.

8. Cross-track contribution and HC-POD integration record

Item	Record
Tracks	Data Science and Project Management.
Dependency	Validated findings and data limits that shape modelling decisions, and issues that shape project decisions.
Output provided	Six validated facts (33% emergency share, 5,000 of 60,000 appointments, unusable phone column, insurance IDs without an uninsured value, department and facility mismatch, unreliable billing status) and eight issues for the log.
Output received	None recorded at the time of writing.
Final integration activity	The facts and issues are the analytics input to the final walkthrough (problem, data, analytics) and to the limitations section.
What changed	The staffing recommendation became conditional and the satisfaction-based recommendation was withdrawn, so other tracks do not build on conclusions that failed testing.
How it improved the solution	Downstream work starts from tested numbers and from a documented list of data limits.
Evidence	Test results table, robustness outputs, dashboard exports, Week 7 report.
Contribution to the presentation	The analytics section, from the data problem to the validated findings and recommendations.

9. Limitations and open items

- **Data limits.** 8% appointment coverage, contradictory billing status, revenue summary that does not reconcile, corrupted phone numbers, department and facility mismatch, no admissions, laboratory, prescription or referral records, no facility names.
- **Analytical limits.** Facility results rest on small samples (72 to 102 reviews per facility) and are descriptive. Statistical tests were computed from query outputs, not stored in the database.
- **Open dashboard items.** Province map placed outside Rwanda, low contrast in dark matrix cells, ICU label casing.
- **Evidence for the claims above.** Scripts 01 to 04 and the tables clean.dq_report and clean.test_results, the three dashboard exports, the Week 6 insights report and the Week 7 testing report, all in the project repository.